

JEE MAIN REVISION MASTER: SEQUENCES AND SERIES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 13 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Arithmetic & Geometric Progressions (AP, GP, AGP)	12 PYQs	Progressions
Standard	AM-GM-HM Inequality & Telescoping Series	1 PYQs	Inequalities & Sigma

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Arithmetic & Geometric Progressions (AP, GP, AGP)	$S_n = \frac{n}{2}[2a + (n-1)d]$, $S_n = \frac{a(1-r^n)}{1-r}$, $S_{\infty} = \frac{a}{1-r}$; $(r < 1)$, $T_n = S_n - S_{n-1}$
AM-GM-HM Inequality & Telescoping Series	$\frac{a+b}{2} \geq \sqrt{ab} \geq \frac{2ab}{a+b}$, $\sum n^2 = \frac{n(n+1)(2n+1)}{6}$, $\sum n^3 = [\sum n]^2$, $\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Arithmetic & Geometric Progressions (AP, GP, AGP) (High Yield — 12 PYQs)

Core Principle: nth term and sum formulas for AP and GP, infinite geometric series convergence, and Arithmetico-Geometric series summation.

■■ **Caution / JEE Pitfall:** The formula $S_{\infty} = \frac{a}{1-r}$ is valid ONLY when $|r| < 1$. If $|r| \geq 1$, the infinite series diverges.

■ **Speed Trick:** To choose 3 terms in AP: $a-d, a, a+d$ (sum eliminates d). In GP: $a/r, a, ar$ (product eliminates r).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "If we insert two numbers between $\sqrt{2}$ and 4 so that the resulting sequence is in G.P, then the inserted numbers in the order are..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "If 4th, 10th and 16th terms of a G.P. are x, y and z respectively, then..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "If $S_n(n)$ stands for sum to n -terms of a GP with a as the first term and r as the common ratio, then $S_n(n):S_{2n}$ is..." [Ans: N/A]

3.2 AM-GM-HM Inequality & Telescoping Series (Standard Yield — 1 PYQs)

Core Principle: Arithmetic, geometric, and harmonic mean inequalities for positive numbers, sum of first n natural powers, and partial fraction telescoping.

■ **Caution / JEE Pitfall:** $AM \geq GM$ equality holds IF AND ONLY IF all participating positive numbers are equal ($a_1 = a_2 = \dots = a_n$).

■ **Speed Trick:** Whenever asked for the minimum value of a sum of positive terms whose product is constant (e.g. $x + (1/x)$), use $AM \geq GM$.

Representative Source PYQs:

- [2022 Shift] Q#2 (MCQ): "If $A=1,2,3,\dots,10$, then number of subsets of A containing only odd numbers is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

If we insert two numbers between $\sqrt{2}$ and 4 so that the resulting sequence is in G.P, then the inserted numbers in the order are

- (A) $8, \sqrt{2}$
- (B) $2, \sqrt{8}$
- (C) $\sqrt{8}, 2$
- (D) $\sqrt{2}, 8$

Correct Answer: None

Solution Analysis: The sequence after inserting two numbers will be $\sqrt{2}, x, y, 4$. Since these four terms are in G.P., let the common ratio be r . The first term is $\sqrt{2}$ and the fourth term is 4. For a G.P., the fourth term is ar^3 . So, $\sqrt{2}r^3 = 4$
 $r^3 = \frac{4}{\sqrt{2}}$ Now, $\frac{4}{\sqrt{2}} = \frac{2^2}{2^{1/2}} = 2^{3/2}$ So, $r^3 = 2^{3/2}$ Taking cube root, $r = 2^{1/2} = \sqrt{2}$ Now the terms are:
 $\sqrt{2}, \sqrt{2} \times \sqrt{2}, 2 \times \sqrt{2}, 4$ So, $x=2$ and $y=\sqrt{8}$ Hence, the inserted numbers in order are $2, \sqrt{8}$ Correct option: Option B.

Benchmark Problem 2: KCET 2022 (Q#2)

If $A=1,2,3,\dots,10$, then number of subsets of A containing only odd numbers is

- (A) 31
- (B) 27
- (C) 32
- (D) 30

Correct Answer: None

Solution Analysis: Given, $A=1,2,3,\dots,10$ Set A containing only odd number $=1,3,5,7,9$ Number of subsets $=2^5$ [it contains empty set also] The required number of subsets of A containing only odd number $=2^5 - 1 = 32 - 1 = 31$